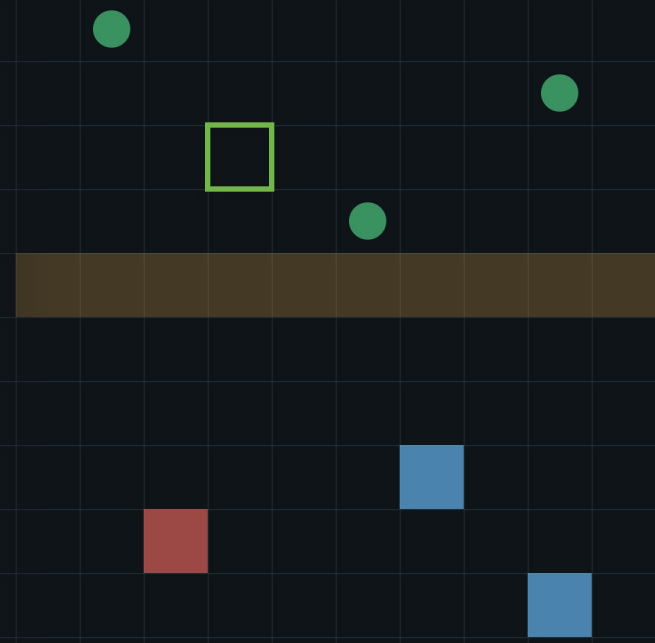
OPEN AGENT HACKATHON 2026 · TRACK 3 REASONING ARCHITECTURE

Shop-Floor OS

AI agents plan. Robots execute. Humans judge.

A multi-agent control room for a car final-assembly hall, reasoning with NVIDIA Nemotron through Meterless Nightshift, on plant data federated by Zetaris.



Team Spartans · Hemanth Raja Guttikonda · Poojitha Katta · October 2026

THE PROBLEM

A line stop still gets a radio call

$2.3M

lost per hour of unplanned downtime in an automotive plant

Slow, manual response

A station runs short, someone pulls the andon cord, a team leader radios for parts and waits.

Robots and people share the floor

One rushed AGV dispatch through a pedestrian crossing turns a delay into a safety incident.

Nothing is learned

Fleet, people, machines and history sit in separate systems, so the next stop starts from zero.

Source: Siemens, “The True Cost of Downtime 2024”.

WHO IT IS FOR

Built for the shift supervisor

TODAY

- Four systems to check: AGV fleet, staff and certificates, machines, incident history
- Decisions made by radio, with no record of why
- Safety rules live in binders, not in the dispatch

WITH SHOP-FLOOR OS

- One screen: the hall map, open incidents and a decision panel
- Agents plan; routine safe actions run, risky ones wait for approval
- Every plan is checked against OSHA clauses and logged with its reasons

Also served: team leaders, technicians and material handlers, who receive the dispatched tasks.

THE SOLUTION

One control room, four agent teams

Incident response

Fences the area, plans who and which AGV, checks safety, dry-runs, dispatches, verifies.

Autopilot

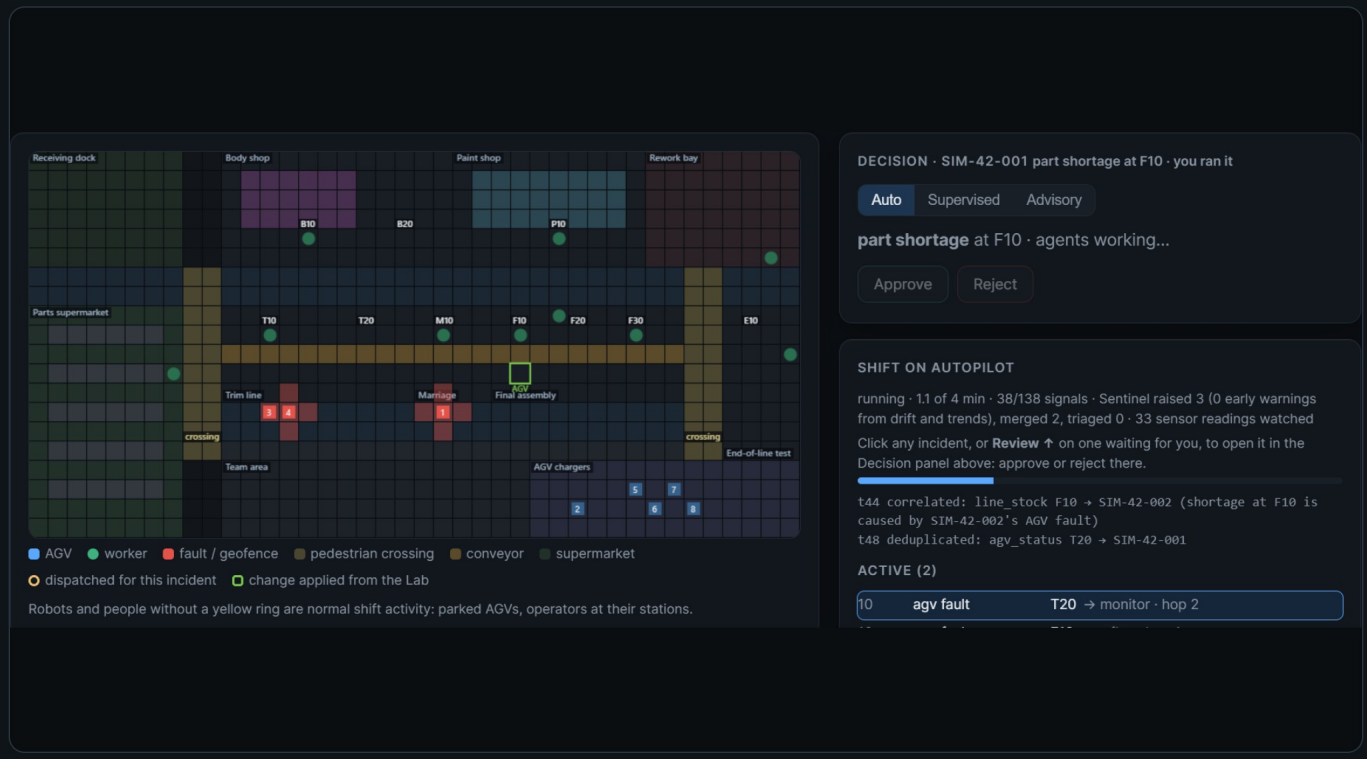
The Sentinel watches every signal and triages incidents on its own.

Ask the floor

Plain-language questions, answered from Zetaris data with every claim cited.

Lab

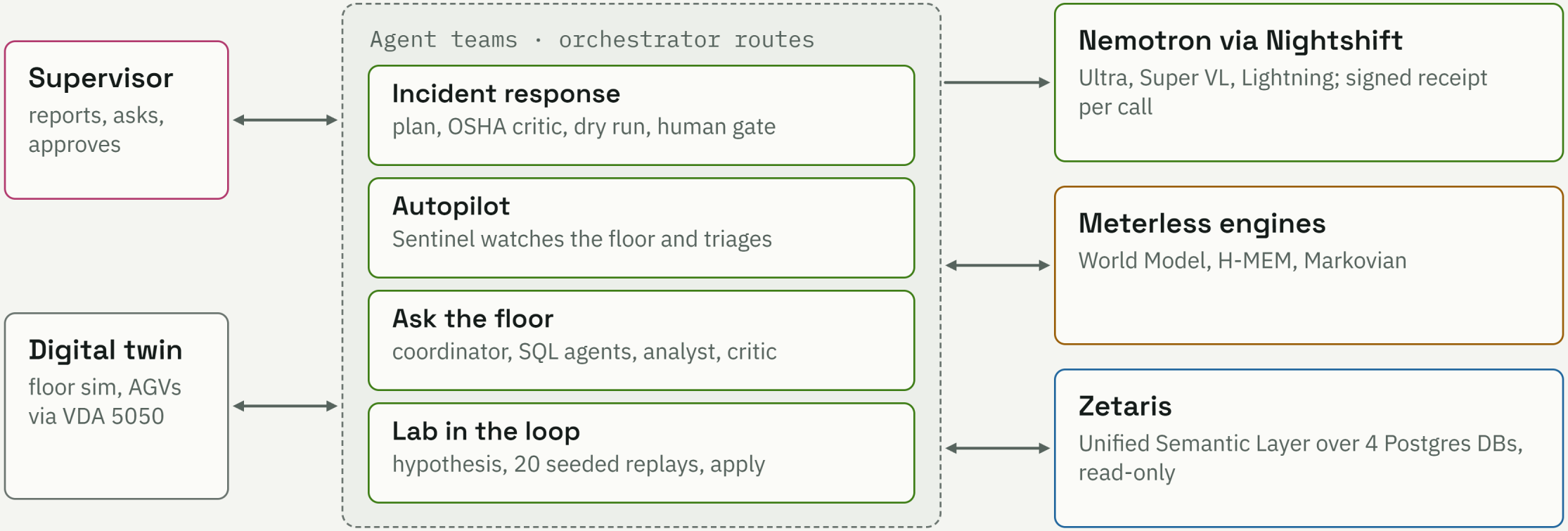
Tests one floor change against 90 days of history before anyone moves a thing.



Fictional car final-assembly hall: 10 stations, 8 AGVs, 24 workers, 2,071 past incidents.

ARCHITECTURE

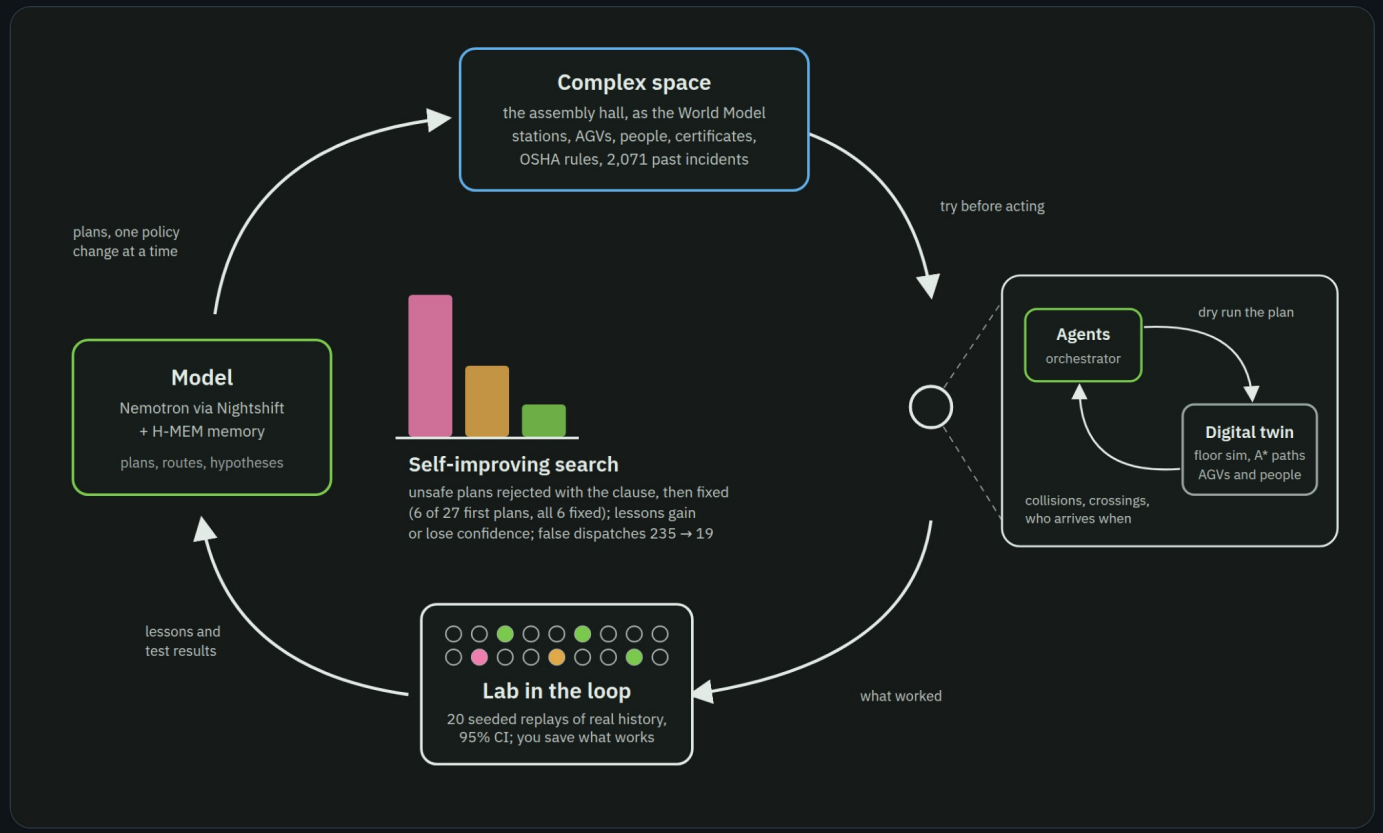
How it fits together



Border colour: green calls Nemotron, amber Meterless engine, blue Zetaris, pink a person, grey code.

LAB IN THE LOOP · DIGITAL TWIN

Nothing moves until the twin has tried it



Try before acting

Every plan is dry-run in the twin: collisions, crossings, who arrives when

Test on history

The Lab replays real incidents, 20 seeds, with a 95% confidence interval

It gets better

6 of 27 first plans were unsafe, all fixed; false dispatches fell 235 → 19

Every plan is dry-run in the twin before dispatch. No unsafe plan has been executed.

HOW AN INCIDENT RUNS

A safety chain the model cannot skip

01
Report
you, or the Sentinel

02
Reflex
fence the zone first

03
Monitor
real, or noise?

04
Candidates
certified people, free AGVs

05
Planner
who does what

06
Safety critic
cites OSHA clauses

07
Dry run
plan tried in the twin

08
Human gate
act, or ask you

09
Executor
dispatch on A* routes

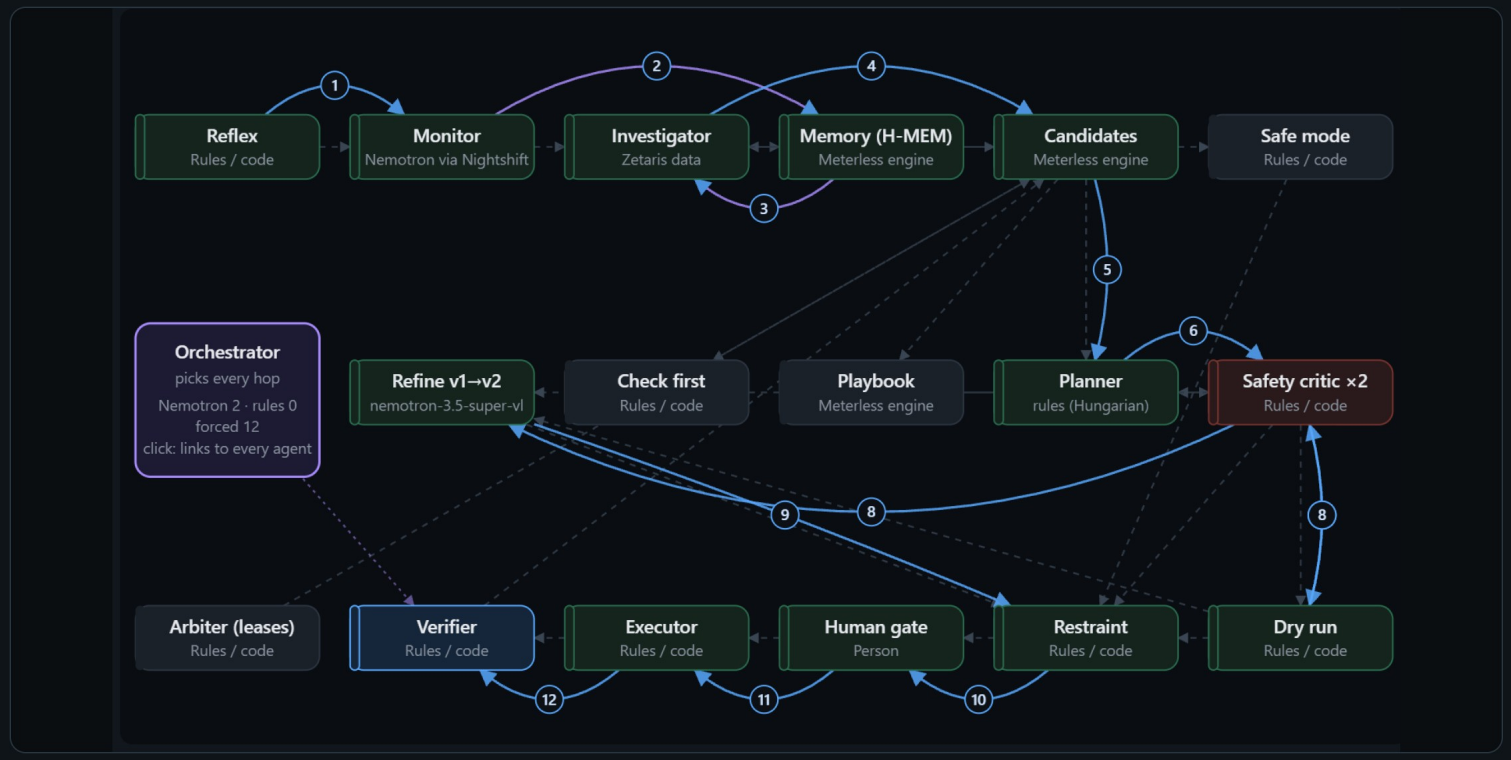
10
Verifier
re-plan if stuck

An **orchestrator** picks each next hop with Nemotron Lightning; code decides which hops are legal, so critic, dry run and gate always run. Every incident ends as **ACT**, **VERIFY**, **SAFE-HOLD** or **ESCALATE**.

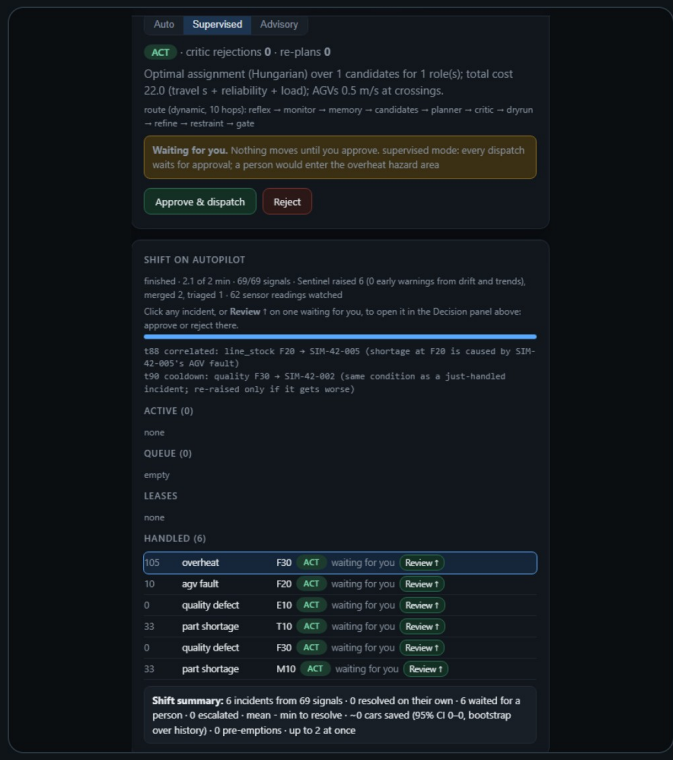
Nemotron via Nightshift | Meterless engine | code, no model | a person decides

TRACEABLE BY DESIGN

Every hop, and who chose it, is on screen



A live incident: Nemotron Super refines the plan while the checked rules plan is already safe.



Human gate: nothing moves until you approve.

NVIDIA NEMOTRON THROUGH METERLESS NIGHTSHIFT

Nemotron reasons in every agent team

`nemotron-3-ultra-550b-a55b`

Reasons hardest

First choice for the planner, analyst and Lab hypotheses

`nemotron-3.5-super-vl-120b-a12b`

Thinks deeply

Coordinator, SQL agents, shift handover, zone briefing; backs up Ultra

`nemotron-3.5-lightning-30b-a3b`

Decides fast

Router, Monitor, Sentinel triage, evidence critic; last fallback everywhere

6,034 / 6,034

Nightshift compute receipts verified (Ed25519)

- Nightshift is the only AI provider
- Three Nightshift apps, one per model tier
- Fallback chain per agent role
- Model named on every trace line

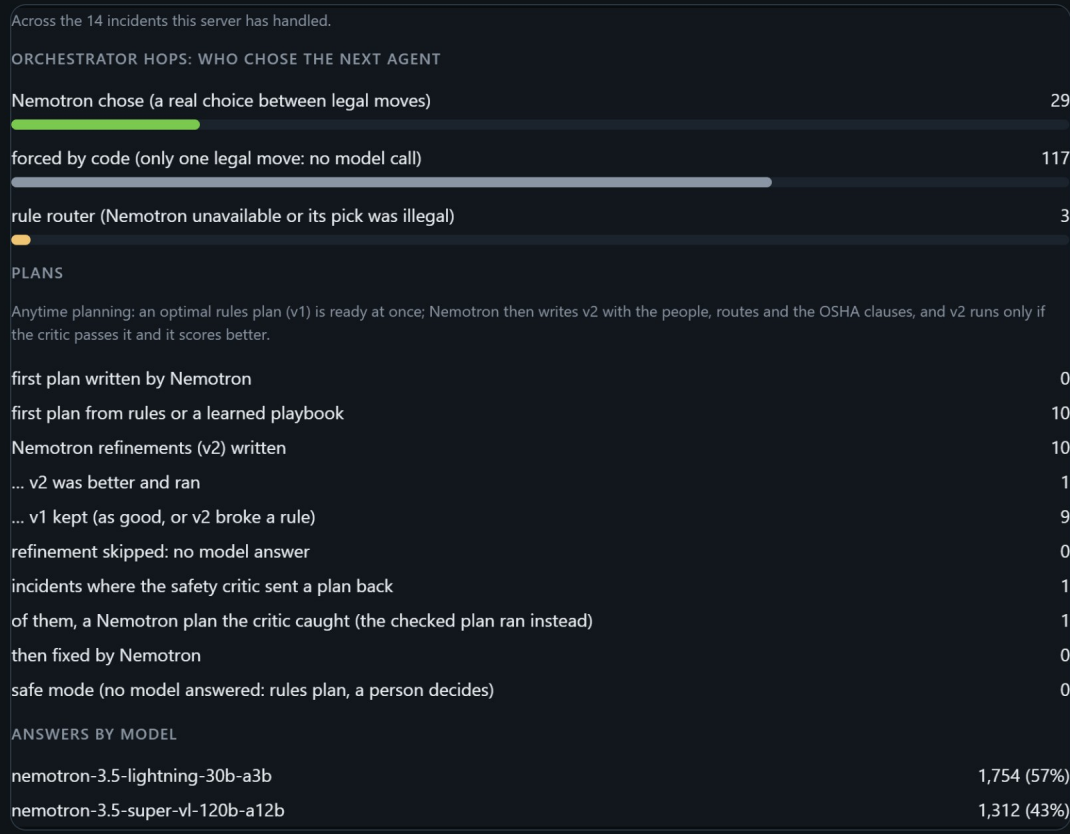
Ultra was pinned on the last day; the evaluation numbers in this deck were measured with Super VL as the planner.

NEMOTRON AT WORK, MEASURED

We count what Nemotron decides

Same live shift, seed 31	Before	After
Incidents in safe mode	13 of 19	0 of 14
Hops Nemotron chose	6	29
Hops the rule router took	50	3
Nemotron refinements	0	10

The “Nemotron at work” tab showed that under load most incidents never reached the model. Three fixes: larger token budgets for Lightning, queue timeouts no longer counted as model failures, and one circuit breaker per role.



Source: “Nemotron at work” tab on the live server, 25 Oct 2026. Chaos suite re-run after the fixes: 15 of 15 pass.

CRITIQUE AND RETRY

Nemotron proposes. Code checks.

216/216
planted unsafe plans caught by the critic

6 of 27
live first plans were unsafe; all 6 fixed from the critic's feedback

0
unsafe plans executed

9/9
reruns gave the same decision

Critic rejection, cited to the clause · 29 CFR 1910.178(n)(4)

"The driver shall be required to slow down and sound the horn at cross aisles and other locations where vision is obstructed."

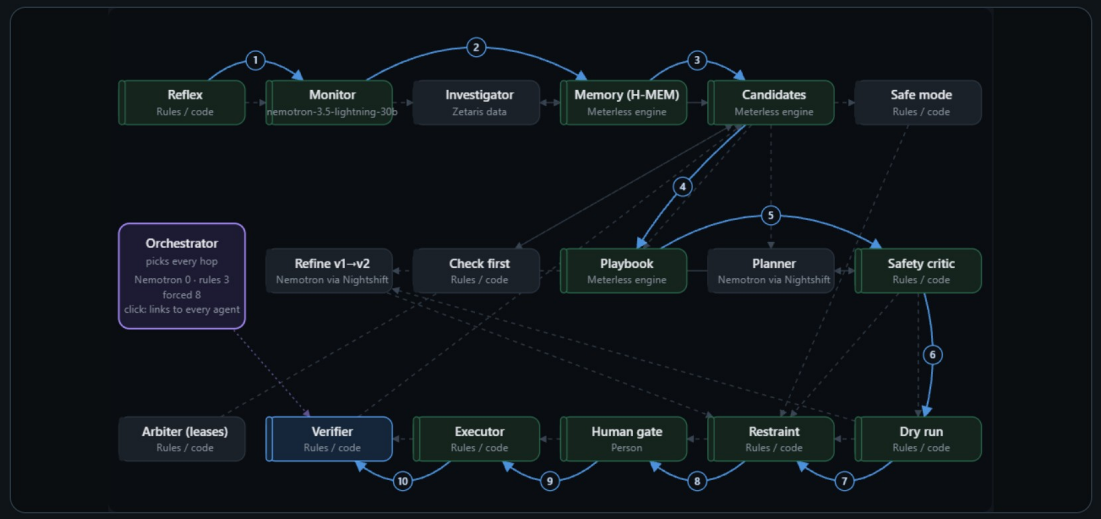
A plan that sends an AGV through a crossing at full speed is rejected with this clause. The planner gets the clause back and re-plans at 0.5 m/s.

From python run.py eval --live, with Nightshift answer memory switched off.

TWO LEARNING LOOPS, LIVE

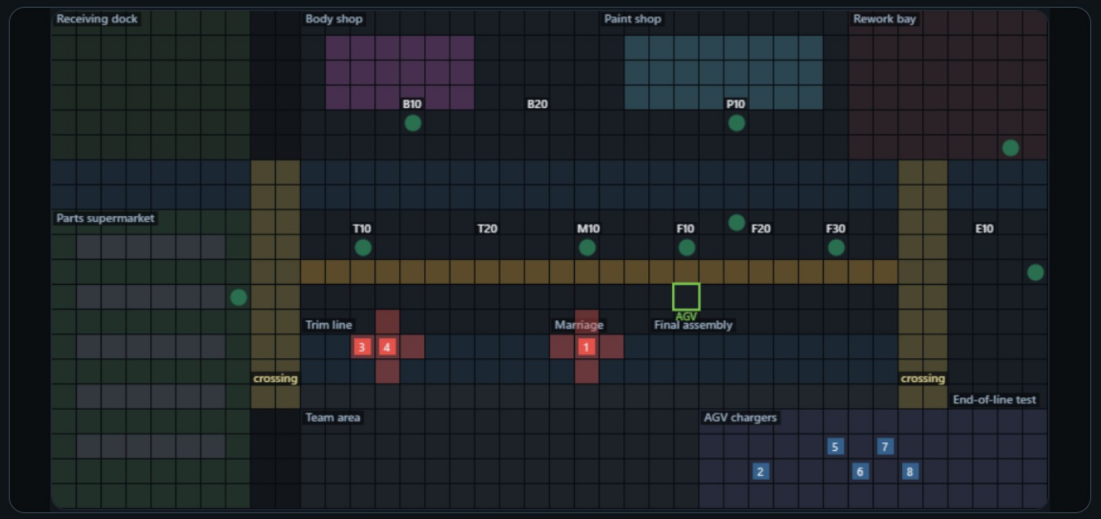
It learns at two speeds

FAST LOOP · EVERY INCIDENT



SIM-42-1634, part shortage at F20. H-MEM recalled the playbook from earlier F20 shortages (hop 4). The plan still went through the critic and dry run, and the outcome updates the lesson's confidence.

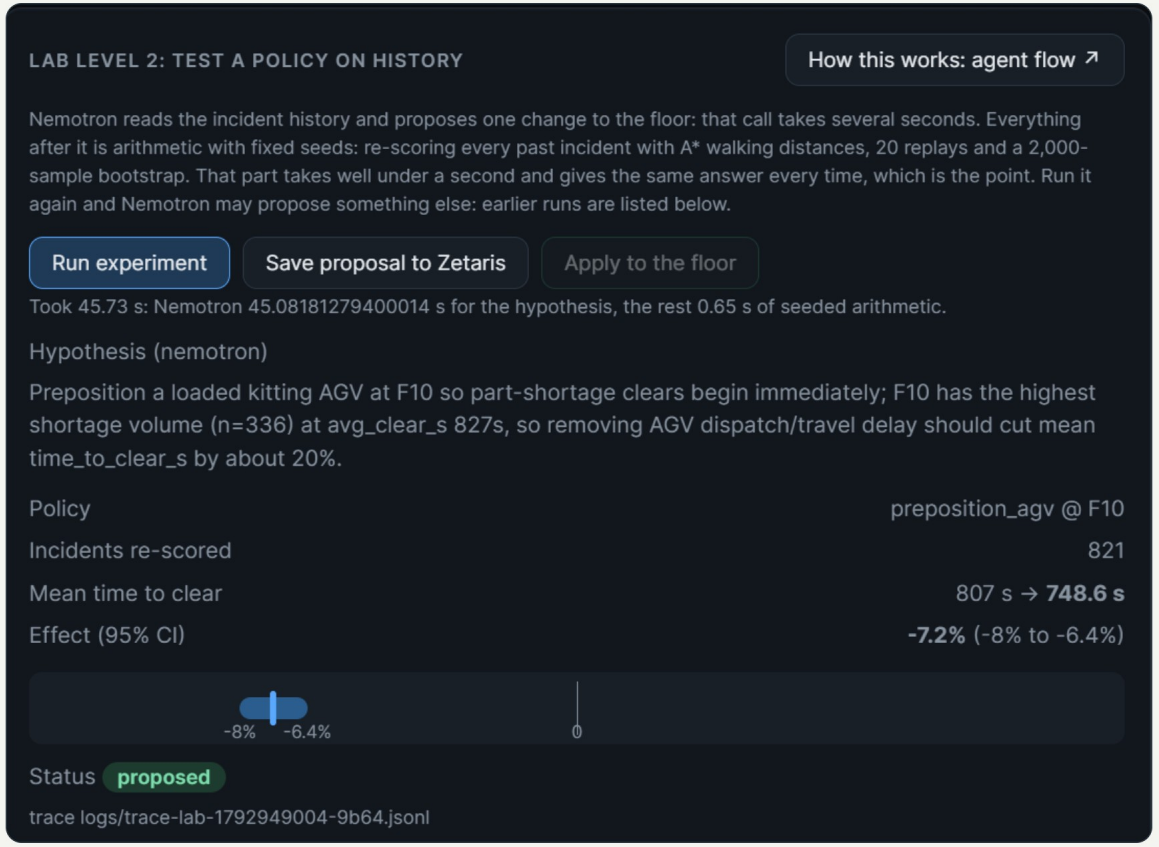
SLOW LOOP · THE LAB



Kitting AGV waiting at F10. Nemotron proposed it, the Lab tested it on 821 past incidents (−7.2%), and it was applied: the green square. Each new F10 shortage is measured against the prediction.

LAB IN THE LOOP

Test a floor change on history first



Nemotron's hypothesis

"Preposition a loaded kitting AGV at F10 so part-shortage clears begin immediately."

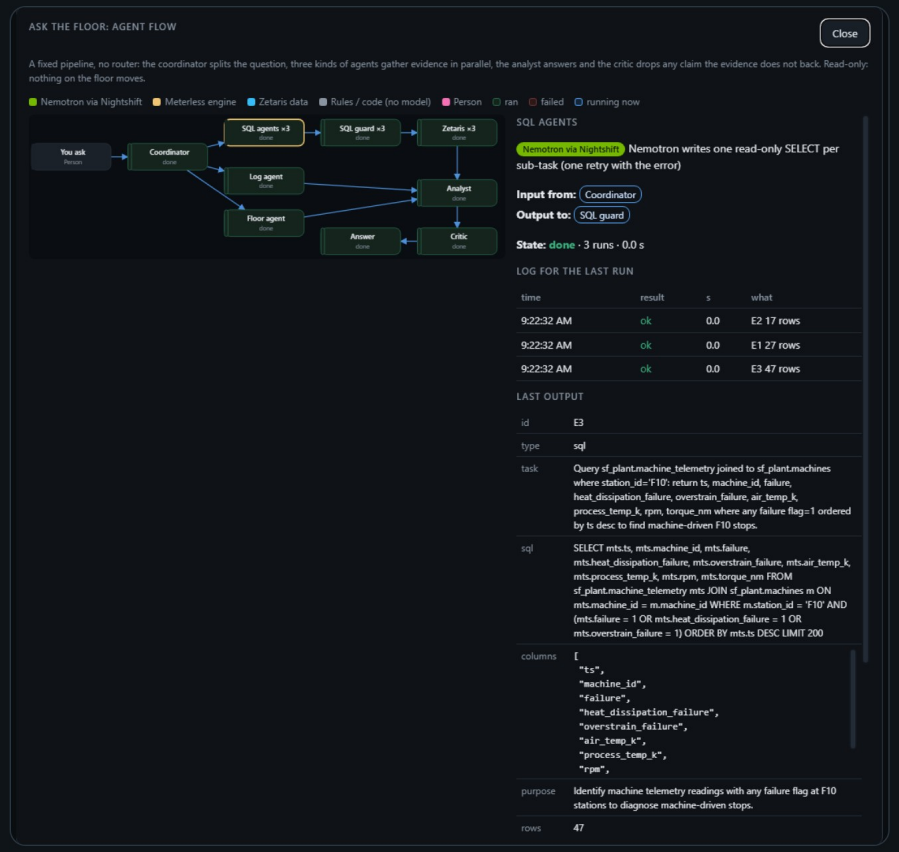
−7.2%

time to clear, 95% CI −8.0% to −6.4%

821 past incidents re-scored with A* walks: 807 s → 748.6 s. 45 s of Nemotron, then 0.65 s of seeded arithmetic.

ASK THE FLOOR

Answers that cite their rows



"Why does F10 keep stopping?"

Coordinator splits the question into sub-tasks

SQL agents write one SELECT each, through a SQL guard, under a read-only Zetaris role

Log and floor agents add H-MEM lessons and live World Model facts

Analyst answers; the critic drops any claim the evidence does not back

RELIABILITY

Built to fail safe, and tested that way

15/15

chaos scenarios pass: Nightshift down, 60% overload, broken JSON, a swapped model, Zetaris down

19 at once

incidents on models failing 30% of calls: all 19 decided, 0 unsafe

313

automated tests pass, including Meterless engine conformance

Anytime planning

An optimal rules plan is ready first; Nemotron’s plan replaces it only if safer and better.

Read-only by construction

Agents query Zetaris with their own login; an INSERT is refused.

Never the unsafe way

No answer from a person: remind at 2 min, escalate at 4, hold hazards safe at 6.

Monitor: 19 false alarms sent to the floor vs 235 for a threshold alarm. Critic: 0 of 54 safe plans wrongly rejected.

TECHNICAL STACK

What it runs on

Layer	Technology
Models	NVIDIA Nemotron 3 Ultra 550B, Nemotron 3.5 Super VL 120B and Lightning 30B, through Meterless Nightshift with the OpenAI Python SDK; reasoning off, medium or full per role
Data	Zetaris over four Postgres databases in Docker, one Unified Semantic Layer (9 entities); agents use a read-only role; agent logs sync back every 30 s
Agents	Python 3.11, Pydantic contracts between agents; Meterless World Model, H-MEM, Markovian
Server and UI	Standard-library HTTP server with Server-Sent Events; vanilla JS, canvas map, SVG graphs
Floor and robots	Seeded 20×40 grid, space-time A* with reservations, Hungarian assignment, VDA 5050 v2.0 orders
Operations	Dockerfile, JSONL traces, hash-chained journal, restart-safe state, SAMPLE_MODE, 313 pytest tests

Built with Cursor (the model layer, app/llm.py and its tests) and Claude Code (most of the rest).

SPONSOR TECHNOLOGY

How Zetaris and Meterless are used

Sponsor	What it does here	Where it shows
NVIDIA Nemotron	Ultra, Super VL and Lightning plan, route, monitor, query, analyse and propose	Model on every trace line
Meterless Nightshift	The only AI provider: NVIDIA lane, fallback chains, signed receipts	Live receipt strip, Models panel
Meterless engines	World Model grounds plans; H-MEM keeps lessons; Markovian handover uses 67% fewer prompt tokens	What the agents know; conformance tests
Zetaris	Unified Semantic Layer over 4 Postgres DBs; agent logs synced back every 30 s	SQL and rows in every Evidence pane

WHAT THE SPONSOR TECH MADE POSSIBLE

What would be hard without it

NVIDIA Nemotron

Three sizes, one family, one endpoint: Lightning routes every hop, Super VL writes SQL, Ultra plans; each backs up the next.

Otherwise: one big model is too slow per hop; one small model plans poorly.

Meterless Nightshift

A signed receipt for every call and pinned lanes. Every AI decision on the floor is provable, and a swapped model is caught.

Otherwise: the audit trail is only our own logs, which we could edit.

Meterless engines

World Model rejects plans naming people or AGVs that don’t exist; H-MEM turns incidents into playbooks; Markovian keeps handover bounded.

Otherwise: state and history stuffed into prompts that grow and drift.

Zetaris

A Unified Semantic Layer over fleet, HR, plant and ops, queried in place, read-only. Logs are written back, so agents and the Lab query their own history.

Otherwise: a stale ETL copy, and four drivers and credentials inside the agents.

Cursor wrote the model layer and its tests; Claude Code wrote most of the rest.

LIMITS AND FUTURE WORK

Honest limits, a clear next step

STATED LIMITS

- A synthetic hall anchored on public data (AI4I 2020, OSHA, BLS) and real OSHA clauses
- Lab savings are measured in the twin, not on a real floor
- Fleet, MES and PLC connectors are interfaces, not wired to hardware
- Cloud Nemotron only; Ultra pinned on the last day, so evaluations used Super VL

NEXT

- Pilot on one real line through the MES and fleet connectors
- Grow the Zetaris semantic layer beyond nine entities as real systems join
- Nemotron on a DGX Spark beside the line for the fast router
- Share H-MEM lessons across plants; measure Lab changes on real shifts

SHOP-FLOOR OS · TEAM SPARTANS

AI agents plan.
Robots execute.
Humans judge.

Demo video (3:56, recorded live)
youtu.be/YExRZBpAkTA
backup on Google Drive

Project site and replays
hemanth-raja.github.io/shopfloor-os

Code
github.com/Hemanth-Raja/shopfloor-os

Try it without keys
`SAMPLE_MODE=true python run.py run input_examples/example_1.json`